

French Motor Claims Analysis

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Abstract

This report develops and analyzes generalized linear models (GLMs) for the Casualty Actuarial Society's (CAS) French Motor Claims dataset, comprising of a portfolio of over 650,000 policies. After data preprocessing and exploratory data analysis, Poisson, Gamma, Negative Binomial, and Tweedie GLMs were fitted to model claim frequency, severity, and pure premium. Considerations such as AIC, deviance, and overdispersion were used to compare the Poisson and Negative Binomial frequency models. Each GLM coefficient estimates were used to compute relativities and rating tables for certain categorical rating factors; BonusMalus proved to be the strongest rating factor with a Poisson relativity of 4.95 for the 101-150 level and Gamma relativity of 1.84 for the 91-100, the largest relativity in their respective models. Model performance was assessed using a goodness-of-fit statistics, validation set, residual diagnostics, lift charts, ordered Lorenz curves, and Gini coefficients. The Frequency $\times$ Severity and Tweedie model performed similarly in predictive power and risk segmentation, with the Tweedie model maintaining a slight advantage (53 less MAE and 0.01 larger Gini coefficient). Between the two models, the apt model varies on intent. Frequency $\times$ Severity is useful for quantifying the effects of rating factors on claim frequency and severity separately; the Tweedie model can predict the pure premium directly, and has slightly better predictability in this case. Because their performances are close, the correct model is left for interpretation; both can have an advantage depending on the business scenario.

For the GitHub documentation used to produce all outputs and figures, follow the instructions on <https://github.com/ryangomberg/France-Auto-Insurance-Modeling>.

Table of contents

Data Overview and Preprocessing	4
Exploratory Data Analysis	8
Claim Frequency: Distribution and Univariate Analysis	8
Claim Severity Distribution	11
The BonusMalus Feature	12
Modeling Preparation	13
Methodology	14
Poisson GLM	14
Negative Binomial GLM	15
Gamma GLM	15
Tweedie Distribution	15
Model Comparison	16
Frequency Modeling	16
Poisson GLM, Log-Link	17
Negative Binomial, Log-Link	19
Severity Modeling	21
Gamma GLM, Log-Link	21
Tweedie Distribution	23
Pure Premium: Relativity and Comparison	23
Rating Tables	25
Model Diagnostics	28
Residuals	28
Lift Charts	31
Lorenz Curves and Gini Coefficients	32
Model Comparison and Summary	34
Appendix	37
Frequency Modeling	37
Poisson GLM Summary	37
Poisson GLM Relativity	38
Negative Binomial GLM Summary	39
Severity Modeling	40
Gamma GLM Summary	40

Gamma GLM Relativity Summary	42
Pure Premium: Relativity and Comparison	43
Relativity Comparison Table	43

Data Overview and Preprocessing

This dataset was taken from Kaggle (Source: <https://www.kaggle.com/datasets/floser/french-motor-claims-datasets-fremtpl2freq>). Risk factors were gathered from 678,013 third-party motor liability policies from France and consolidated into a frequency table. A severity table is also available to represent the magnitude of a claim. In total, there are 26,639 individual claims.

The frequency and severity tables were joined by the driver's policy ID through PowerQuery. If a driver has no claims, the corresponding entry in the Claim.Amount column is 0.

We present a brief summary of the features that shape each policy. The rating factor's alias in the dataset is written in orange:

- **IDpol**: Policy ID for the driver
- **ClaimNb**: Number of claims during the exposure period
- **Exposure**: Exposure period
- Geographic features, including: Area Code (**Area**), **Density** of inhabitants (number of inhabitants per km²) in the city the driver of the car lives in (**Density**), and the policy regions in France (**Region**). Note that **Area** ranges from rural areas (A) to urban (F), with suburban lying in the middle (C, D)
- Vehicle information, including: The power of the car, ordered categorically (**VehPower**), the age of the vehicle, in years (**VehAge**), the car brand (**VehBrand**), and the car gas (**VehGas**), expressed as Regular or Diesel
- **DrivAge**: The driver's age in years
- **BonusMalus**: In France, <100 and >100 implies bonus and malus, respectively.
- **Claim.Amount**: The amount of a claim linked to a policy, if one was filed.

Some categorical features may have a plethora of classes, such as vehicle brands and regions. Rather than looking through the dataset, we can extract the list of unique classes directly.

List of Area Types: D B E C F A

List of Vehicle Brands: B12 B6 B3 B2 B5 B10 B14 B13 B4 B1 B11

List of Gas Types: Regular Diesel

List of Regions: R82 R22 R72 R31 R91 R52 R93 R11 R24 R94 R83 R54 R26 R53
R73 R42 R25 R21 R41 R43 R74 R23

While all of the data cleaning was done in Excel and PowerQuery, we should double check for any missing entries within the dataset.

	IDpol	ClaimNb	Exposure	Area	VehPower	VehAge	DrivAge	BonusMalus	VehBrand
1	1	1	0.10	D	5	0	55	50	B12
2	3	1	0.77	D	5	0	55	50	B12
3	5	1	0.75	B	6	2	52	50	B12
4	10	1	0.09	B	7	0	46	50	B12
5	11	1	0.84	B	7	0	46	50	B12
	VehGas	Density	Region	Claim.Amount					
1	Regular	1217	R82	0					
2	Regular	1217	R82	0					
3	Diesel	54	R22	0					
4	Diesel	76	R72	0					
5	Diesel	76	R72	0					

Figure 1: First 5 entries of CAS French Motor Claims Dataset

	Feature	Missing
IDpol	IDpol	0
ClaimNb	ClaimNb	0
Exposure	Exposure	0
Area	Area	0
VehPower	VehPower	0
VehAge	VehAge	0
DrivAge	DrivAge	0
BonusMalus	BonusMalus	0
VehBrand	VehBrand	0
VehGas	VehGas	0
Density	Density	0
Region	Region	0
Claim.Amount	Claim.Amount	0

IDpol	ClaimNb	Exposure	Area
Min. : 1	Min. : 0.00000	Min. : 0.002732	Length:678013
1st Qu.:1157951	1st Qu.: 0.00000	1st Qu.:0.180000	Class :character
Median :2272152	Median : 0.00000	Median :0.490000	Mode :character
Mean :2621857	Mean : 0.05325	Mean :0.528750	
3rd Qu.:4046274	3rd Qu.: 0.00000	3rd Qu.:0.990000	
Max. :6114330	Max. :16.00000	Max. :2.010000	
VehPower	VehAge	DrivAge	BonusMalus
Min. : 4.000	Min. : 0.000	Min. : 18.0	Min. : 50.00
1st Qu.: 5.000	1st Qu.: 2.000	1st Qu.: 34.0	1st Qu.: 50.00
Median : 6.000	Median : 6.000	Median : 44.0	Median : 50.00
Mean : 6.455	Mean : 7.044	Mean : 45.5	Mean : 59.76

3rd Qu.: 7.000	3rd Qu.: 11.000	3rd Qu.: 55.0	3rd Qu.: 64.00
Max. :15.000	Max. :100.000	Max. :100.0	Max. :230.00
VehBrand	VehGas	Density	Region
Length:678013	Length:678013	Min. : 1	Length:678013
Class :character	Class :character	1st Qu.: 92	Class :character
Mode :character	Mode :character	Median : 393	Mode :character
		Mean : 1792	
		3rd Qu.: 1658	
		Max. :27000	

Claim.Amount
Min. : 0
1st Qu.: 0
Median : 0
Mean : 82
3rd Qu.: 0
Max. :4075401

Figure 2: General Statistical Summary of the Dataset

It is worth highlighting the following outliers:

- ClaimNb: The maximum claim number is 16, which is completely unreasonable with an exposure between 0 to 2 years. Any observations with ClaimNb > 4 should be removed from the dataset and rendered as anomalies.
- BonusMalus: Typically, policies begin at a Bonus/Malus value of 100. Any BonusMalus value larger than 150 are extreme outliers. Considering that BonusMalus is likely to have a profound impact on the model, such outliers should be removed.

Additionally, we may want to set a reasonable upper limit on ordinary features as well. For instance, vehicles usually do not last longer than 30-35 years, and most people are not driving past 90 years of age. We can remove such entries without making much impact on feature engineering & modeling results.

Lastly, density is a feature that is often right-skewed. We can observe this when the 3rd quartile is smaller than the mean. Therefore, a logarithmic transformation is suitable in these cases.

We verify that the amendments have been applied correctly.

IDpol	ClaimNb	Exposure	Area
Min. : 1	Min. :0.0000	Min. :0.002732	Length:676280
1st Qu.:1158231	1st Qu.:0.0000	1st Qu.:0.170000	Class :character
Median :2272278	Median :0.0000	Median :0.490000	Mode :character

Mean :2622949	Mean :0.0531	Mean :0.528424	
3rd Qu.:4046529	3rd Qu.:0.0000	3rd Qu.:0.990000	
Max. :6114330	Max. :4.0000	Max. :2.010000	
VehPower	VehAge	DrivAge	BonusMalus
Min. : 4.000	Min. : 0.000	Min. :18.00	Min. : 50.00
1st Qu.: 5.000	1st Qu.: 2.000	1st Qu.:34.00	1st Qu.: 50.00
Median : 6.000	Median : 6.000	Median :44.00	Median : 50.00
Mean : 6.453	Mean : 6.989	Mean :45.47	Mean : 59.74
3rd Qu.: 7.000	3rd Qu.:11.000	3rd Qu.:55.00	3rd Qu.: 64.00
Max. :15.000	Max. :30.000	Max. :90.00	Max. :149.00
VehBrand	VehGas	Density	Region
Length:676280	Length:676280	Min. : 1	Length:676280
Class :character	Class :character	1st Qu.: 92	Class :character
Mode :character	Mode :character	Median : 393	Mode :character
		Mean : 1793	
		3rd Qu.: 1658	
		Max. :27000	
Claim.Amount	LogDensity		
Min. : 0	Min. : 0.000		
1st Qu.: 0	1st Qu.: 4.522		
Median : 0	Median : 5.974		
Mean : 82	Mean : 5.983		
3rd Qu.: 0	3rd Qu.: 7.413		
Max. :4075401	Max. :10.204		

Figure 3: General Statistical Summary of the Amended Dataset

In summary, 1,733 entries were removed under these considerations.

Lastly, there are some quantities that are worth considering:

1. Exposure-adjusted frequency: Claim number divided by exposure length.
2. Average claim: For positive claim numbers, it is the claim amount divided by the claim number.
3. Pure premium: Claim amount divided by exposure length.

We will look at some of these quantities on a portfolio-level basis.

Total exposure: 357362.3
Total claim count: 35912
Total claim amount: 55136544
Portfolio frequency: 0.1004919 claims per year

Portfolio average severity: 1535.324

Portfolio pure premium: 154.2875

- Roughly speaking, 1 in 10 policies contain a claim each year.
- When a claim is filed, the mean (or expected) cost is 1,535.32.
- We need to charge 154.29 as the pure premium, before other multipliers are incorporated (expenses, profit margins, etc...)

Exploratory Data Analysis

We are interested in how each rating factor impacts both claim frequency and severity.

Claim Frequency: Distribution and Univariate Analysis

There are two forms of EDA for claim frequency:

1. **Claim count distribution.** In particular, we want to see the number of policies that contain exactly 0, 1, 2, 3, and 4 claims.
2. **Univariate analysis.** More precisely, for some categorical rating factors, we want to see how claim frequency varies over each level. Additionally, it is also worth looking at the policy exposure for each level.

Figure 4 reveals that 642,319, or 94.98%, of policies have zero claims, and that 5.02% of policies have at least one claim. This is hinting at zero-inflated data, where the majority of drivers are claim-free, which is common in insurance modeling. This should be kept in consideration when building frequency and premium models.

It is also worth grouping numerical rating factors with large ranges, such as BonusMalus or DrivAge in particular, for easier analysis moving forward.

Next, we look at how claim frequency changes among multiple rating factors. As shown in Figure 5, the blue line indicates the observed frequency for each class/band and the dashed line represents the portfolio frequency (what we computed earlier). Additionally, exposure is given by the height of each bar.

- Areas E, F (urban) have higher frequencies than A, B (rural). This makes sense, as traffic and accidents are positively correlated.
- Young drivers have a much larger claim frequency. Intuitively, as drivers gain experience, the number of accidents decline and thus claim frequency decreases.

- Bonus/Malus is extremely important here. Poor drivers are penalized, whereas claim-free drivers are rewarded and have a BonusMalus closer to 50.
- Vehicles with more power will typically experience above average claim frequency. This feature does not have a substantial impact on claim frequency.
- Vehicle gasoline does not have a strong influence on claim frequency.

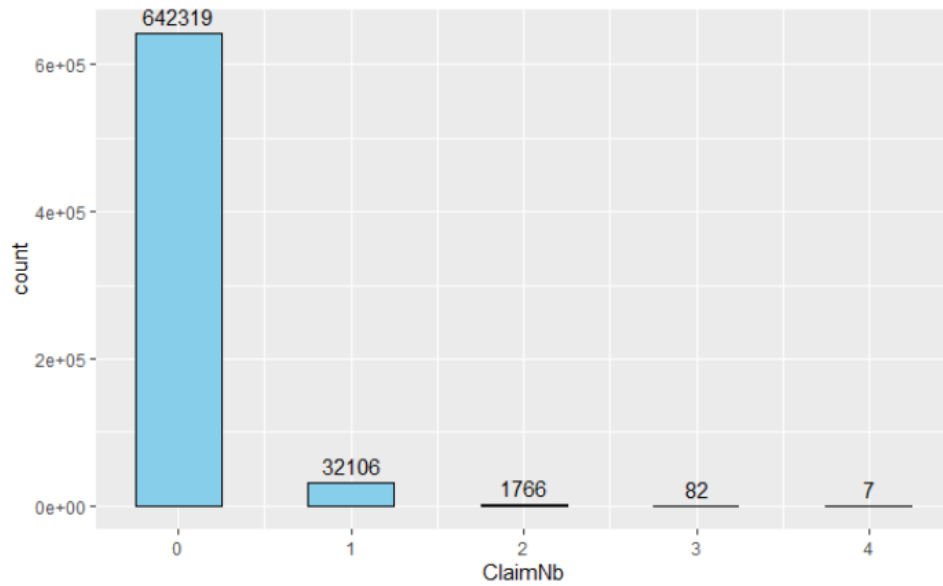
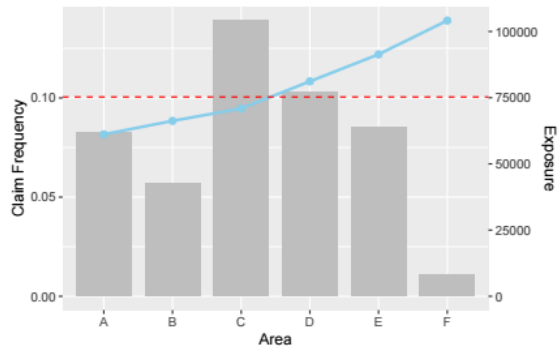
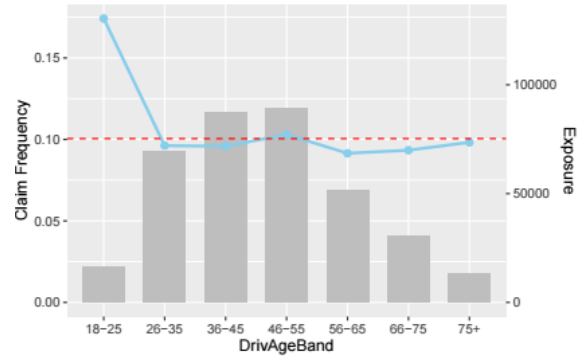


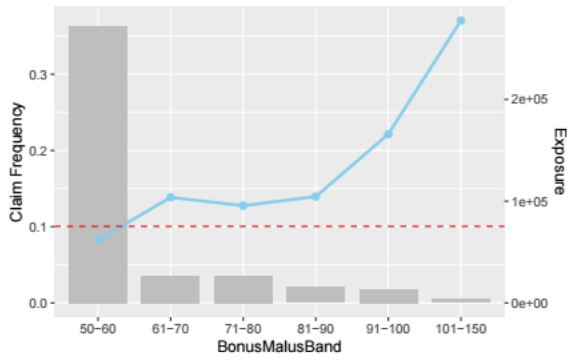
Figure 4: ClaimNb Count Histogram



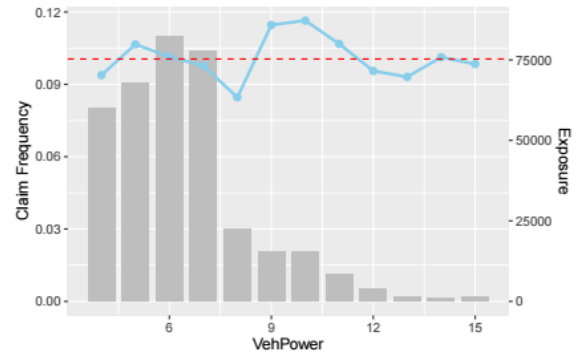
(a) Area



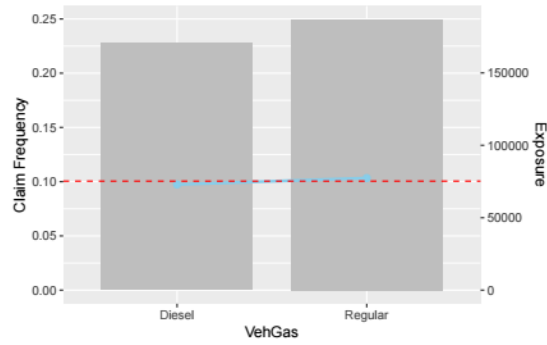
(b) DrivAgeBand



(c) BonusMalusBand



(d) VehPower



(e) VehGas

Figure 5: Univariate Analysis on Claim Frequency and Exposure

Claim Severity Distribution

As with claim frequency, there are two analyses to be done:

1. [Count of Average and Log Severity](#). Since we are dealing with zero-inflated claims data, there should be some skew with the severity distribution.
2. [General Statistics Summary](#). We are interested in the mean/median interaction, as well as the general spread of average claim severity.

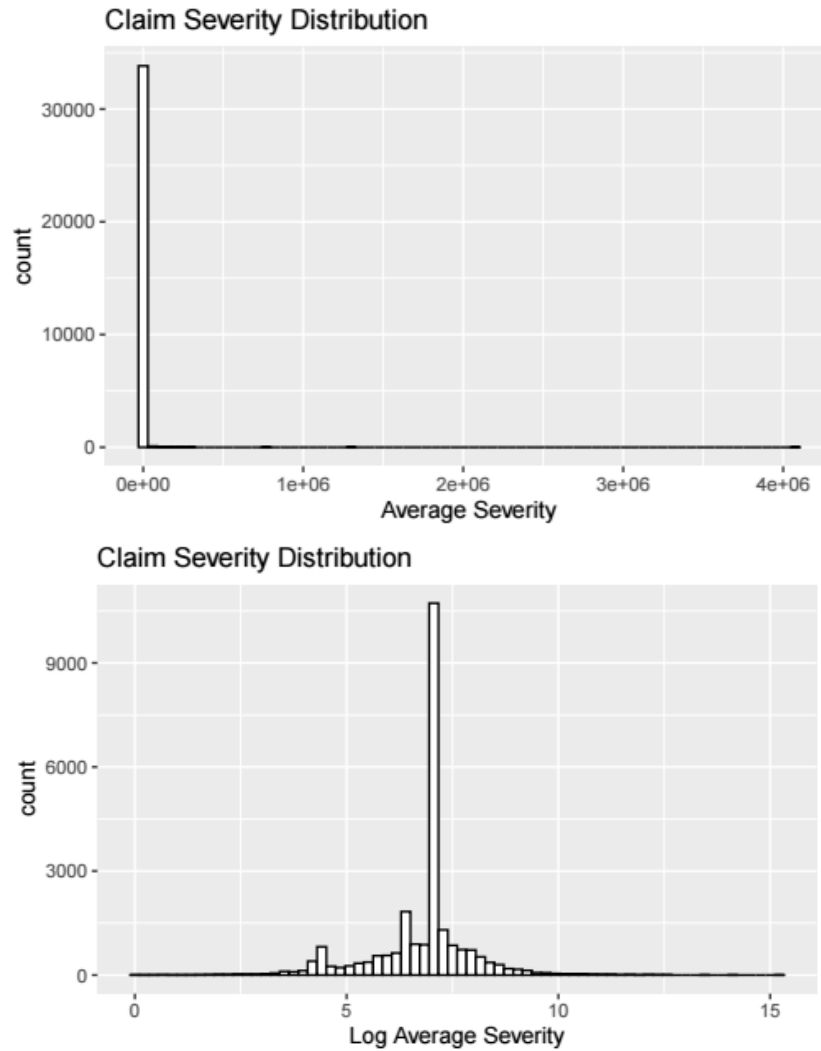


Figure 6: Average and Log Average Claim Severity Count

As observed in Figure 6, there is a strong right-skew and tail. The mean is significantly larger than the median, as a handful of claims are extremely expensive.

Severity Statistics for positive claim frequency:

Mean: 1565.345

Median: 888.99

Max: 4075401

Standard Deviation: 24550.13

The BonusMalus Feature

The BonusMalus Feature is an incredibly important risk factor; it is a strong indicator of claim-free drivers and drivers with claims. Therefore, we look at its count distribution as well as its potential correlation with driving experience.

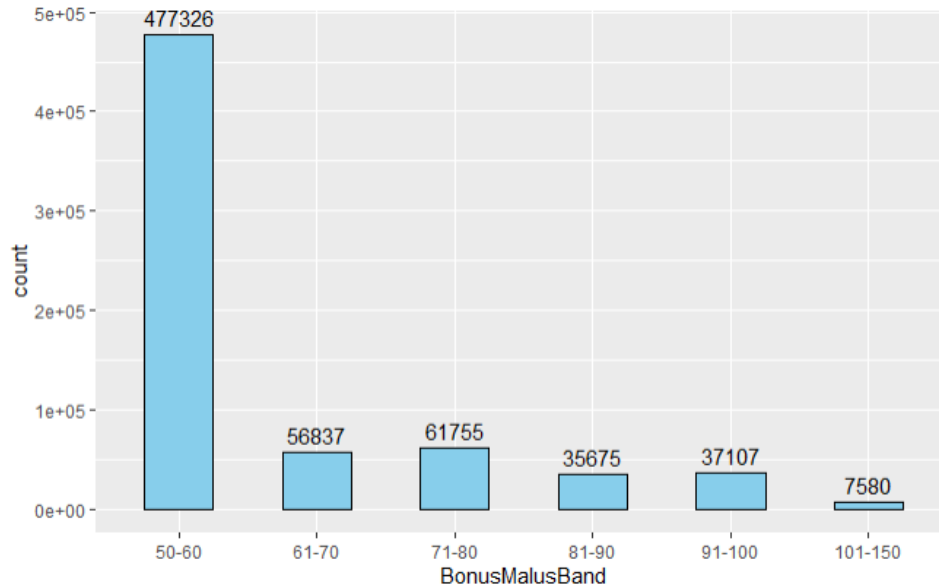


Figure 7: BonusMalusBand count distribution

- Figure 7: A large majority of drivers in this dataset have earned a discount by having no claims.
- Figure 8: Additionally, the relationship between BonusMalus and DrivAge is driven by their interaction, for older drivers have had more time to increase their discount. This can hint at collinearity, and we must keep this interaction in mind during modeling.

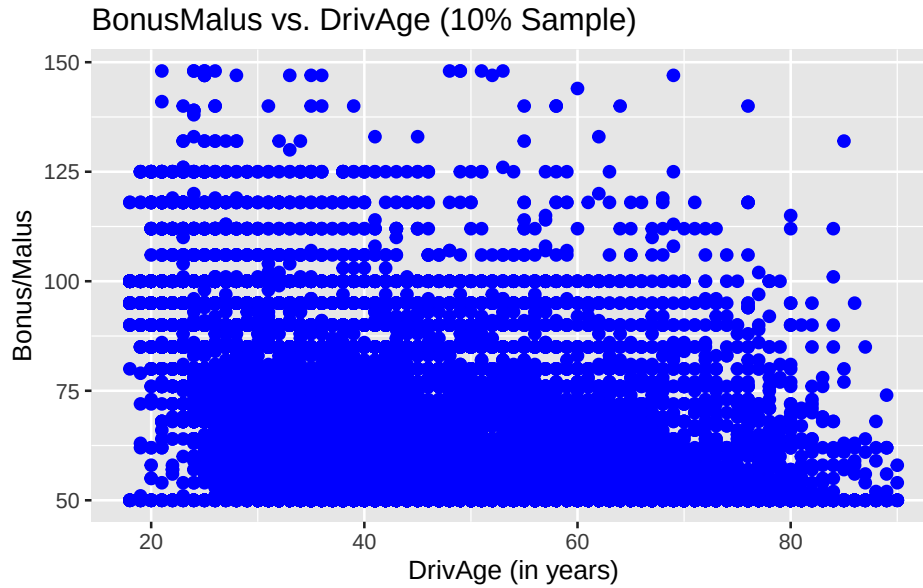


Figure 8: BonusMalus vs. DrivAge Scatterplot

Modeling Preparation

Before proceeding, we condense some discrete features into bands, or risk groups. They allow for easier interpretation and are less sensitive to outliers.

We have already made categorical groups for DrivAge and BonusMalus, and will make groups for VehAge, VehPower, and Density. In particular, Density will be rewritten as an ordinal rating factor, split into quintiles (ex. "Low," ..., "High"). We verify by taking counts per level in VehAge and DrivAge.

	VehAgeBand	n
1	0-1	128988
2	2-4	152786
3	5-7	107216
4	8-10	102888
5	11-20	177219
6	21+	7183

	DrivAgeBand	n
1	18-25	38772
2	26-35	148936
3	36-45	170066

4	46-55	161594
5	56-65	90515
6	66-75	47927
7	75+	18470

The next step is to construct a train-test split (80%/20%), where the claim frequency remains relatively consistent between the training and test set.

Training Claim Frequency: 0.1006647

Test Claim Frequency: 0.09980215

Methodology

Here we present an introduction of the following models (to be used) and their motivation in insurance modeling:

Poisson GLM

Poisson models are an incredibly useful tool for handling counts, which we extend to capturing the number of claims a given policy has. In functional form,

$$E[Y] = \ln(\mu_Y) = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n.$$

A key assumption for Poisson GLMs is that the mean equals the variance, or $\text{Var}(\mu) = \mu$. If the variance exceeds the mean, we have overdispersion. Mathematically,

$$\text{Var}(Y) = \phi\mu, \quad \phi > 1$$

is when overdispersion is present. Poisson models can suffer from this drawback. Ideally, we want the overdispersion parameter to be close to 1. We will check for this later.

While we could predict the raw claim counts, **ClaimNb**, we are far more interested in modeling claim frequency. This can be achieved by dividing the raw claim count over the policy exposure length, giving us a rate of claims per year. This gives us

$$\ln(\mu) \sim \log(\text{Exposure}) + \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n,$$

where the term $\log(\text{Exposure})$ is called an offset.

Negative Binomial GLM

The Negative Binomial model, like Poisson, serves to estimate count and/or frequency. However, the Negative Binomial distribution allows the variance to exceed the mean by adding an extra parameter:

$$\text{Var}(Y) = \mu + \alpha\mu^2.$$

If $\alpha = 0$, the Negative Binomial distributions reduces to the Poisson. In short, Negative Binomial models attempt to resolve the overdispersion that Poisson models are often subject to. We are using both Poisson and Negative Binomial models to model exposure-adjusted claim frequency. Later, model performance metrics used to compare the two will be introduced.

Gamma GLM

Moving onto claim severity, we want to model how much a claim costs, given that a claim was filed. The data is filtered to positive claims and we wish to model the average claim amount. The Gamma distribution is a trivial choice because:

- Its domain lies on $(0, \infty)$, and claim amounts are strictly positive when filtered.
- Variance increases with the mean: $\text{Var}(Y) = \frac{\mu^2}{\nu}$.
- By construction, predictions are always positive.

Additionally, common insurance patterns are naturally accounted for by Gamma models. For instance, expensive claims (ex. sustained injuries) are more variable than smaller incidents. As the variance is squarely proportional to the mean, the variance will be larger when the predicted severity increases.

Tweedie Distribution

In the end, we want to model the pure premium. One of the most popular avenues in doing so is by combining a claim frequency (Poisson, Negative Binomial) and severity (Gamma) model. When multiplying their predicted values, we get the annual pure premium rate (currency amount per exposure unit). An extreme advantage of obtaining the pure premium this way is that we get insight on claim frequency and severity separately.

An elegant alternative to the “Frequency $\times$ Severity” approach is through the Tweedie distribution, which can predict the pure premium in one model instead of two. It is uniquely characterized by the power we raise the mean to, which is set equal to the variance, or

$$\text{Var}(Y) = \phi\mu^p.$$

If $p = 1$ and $p = 2$, we recover the Poisson and Gamma, respectively. When $1 < p < 2$, the Tweedie distribution becomes a blend of the Poisson and Gamma, which is precisely the

distribution of aggregate severity. The unfortunate disadvantage is that we lose the separate frequency and severity patterns that Poisson and Gamma models can provide us with.

Model Comparison

There are two types of model comparisons that we can use for selection: AIC/BIC and Deviance.

AIC, or Akaike's Information Criterion, measures prediction error, and therefore how well a model fits the data. When comparing models, the one with the smallest AIC is typically selected. Note that AIC is "unitless;" a large AIC does not necessarily imply poor model performance, especially if said model has the smallest AIC relative to its competitors. BIC shares the same purpose but AIC, but also penalizes models that use a larger training set.

Deviance is an extremely powerful measurement of how well a GLM fits the data and drives a lot of model comparisons. It measures how much far the fitted model sits from a hypothetical perfect model that reproduces the data exactly. Smaller deviance indicates a better fit.

Lastly, comparisons are only made between GLMs that are predicting the same outcome, or within the "same class." For example, Poisson and Negative Binomial models can be compared directly because they both measure claim frequency, but we could not use AIC/BIC to choose between Poisson vs. Gamma.

Frequency Modeling

Because we are modeling counts, we turn to common tools such as the Poisson and Negative Binomial regression, or GLMs. With categorical risk factors, we assume the first class as the "base case." For example, in [DrivAge](#), the base class is the "18-25" year group.

We use a log-link and take the log of the exposure. We also assume a significance level of 5%. For the full summary of each GLM, refer to the Appendix.

Each GLM will be defined using a log-link, or the logarithmic link function. In summary, a link function is a function of the expected response, μ , which is then related to the linear combination of rating factors (known as the systematic component). For each GLM, we choose to use a log-link, which takes the log of the expected response.

Lastly, each GLM will be used to compute the **relativity**. Because we used a log-link, each coefficient corresponds to a multiplicative effect on the frequency. A rating factor's relativity is obtained by exponentiating its GLM coefficient, or e^β . If the [DrivAgeBand26-35](#) coefficient for the Poisson GLM is 0.20, then its relativity is $e^{0.20} \approx 1.22$. That is to say, 26-35 year-old drivers have 22% higher claim frequency than the reference group, or 18-25 year-old drivers.

Poisson GLM, Log-Link

For the Poisson model, the objective is to:

- Fit the GLM using a log-link with **Exposure** as the offset.
- Compute relativities for each categorical rating factor and compare each level to the base group.
- Check for overdispersion at the 5% significance level.

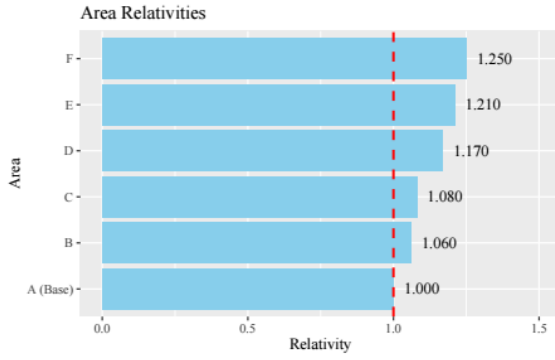
See the Appendix for the full model breakdown. The relativity bar graph for each rating factor is shown on Figure 9 (next page).

As seen in Figure 9, for each graph, the bottom bar corresponds to the base, or “reference group,” whose GLM coefficient is 0. Therefore, its relativity is $e^0 = 1$, and the vertical line serves as this reference point.

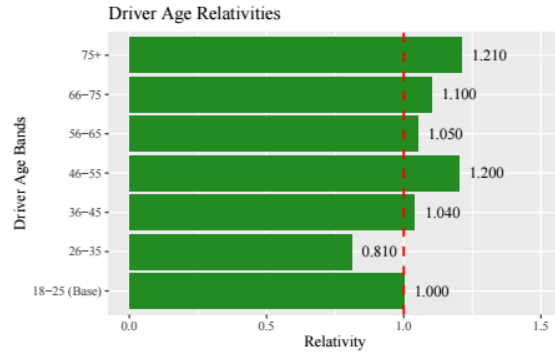
- For rating factor levels whose bar does not exceed the red line, the expected claim frequency is a certain percentage lower than the base level.
- On the other hand, the expected claim frequency is certain percentage higher than the base level if the bar exceeds the vertical line.

Using the relativities for **Area** as an example, all areas (B-F) have an expected claim frequency higher than area A. More precisely, the percentage increase is expected to be 6%, 8%, 17%, 21%, and 25% for areas B-F (compared to area A), respectively.

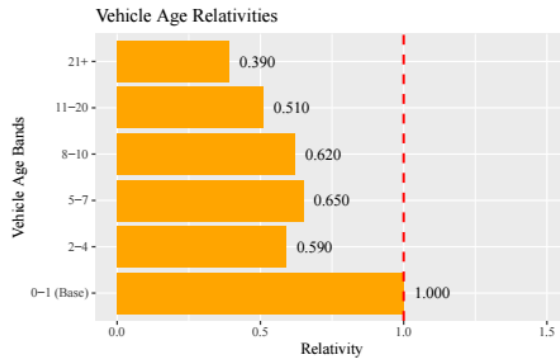
- Older drivers (36+) are subject to a somewhat higher claim frequency than 18-25 years old, up to a 21% increase for drivers of 75+ years of age.
- Any policies with a vehicle age beyond 2 years is expected to have a smaller claim frequency than a new vehicle (0-1 years).
- There is a profound impact of **BonusMalus** on expected claim frequency, especially as a policy’s **BonusMalus** increases. When a policy has claims, **BonusMalus** increases, and subsequently increases the risk of more claims. Policies with a **BonusMalus** between 101-150 have a whopping 495% increase of expected claim frequency with respect to claim-free drivers.
- Vehicles with a power rating of 5-10 have a higher expected claim frequency compared to the base power (4-5).
- Vehicles that use regular gas have an expected 7% increase in claim frequency compared to using diesel.



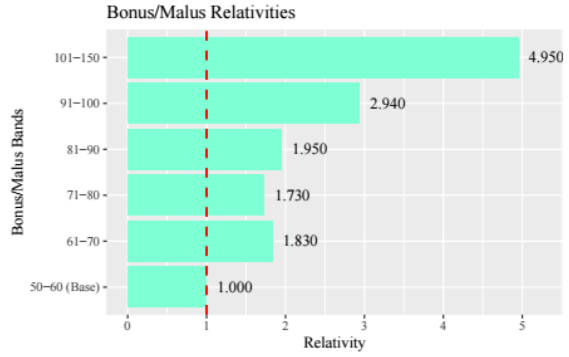
(a) Area



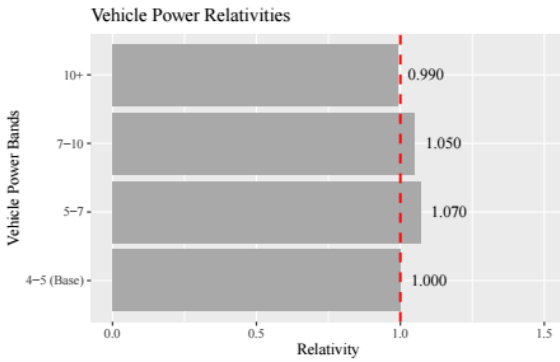
(b) DrivAgeBand



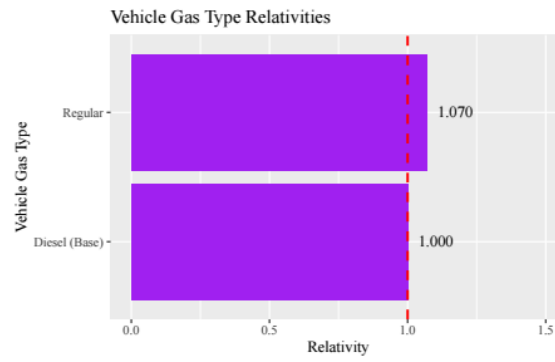
(c) VehAgeBand



(d) BonusMalusBand



(e) VehPowerBand



(f) VehGas

Figure 9: Poisson Frequency Relativities by Level

To better capture this idea, suppose a policy has the following characteristics: Area D, Age 50 years, Vehicle Age of 25, BonusMalus of 50, Vehicle Power of 6, and uses Regular gas. Then

the expected claim frequency is

$$1.17 \times 1.2 \times 0.510 \times 1 \times 1.07 \times 1.07 \approx 0.82,$$

or an 18% decrease in expected claim frequency relative to the base levels. However, this does not necessarily indicate that the severity is also smaller in magnitude compared to the base levels. Some rating factors can have smaller frequency but larger severity (ex. older vehicles cost more to get repaired, but have the lowest claim frequency). This is one such advantage of modeling frequency and severity separately.

Overdispersion test

```
data: FreqPoisson
z = 7.3164, p-value = 1.274e-13
alternative hypothesis: true dispersion is greater than 1
sample estimates:
dispersion
1.171483
```

At a 5% significance level, we can conclude that overdispersion is present in the Poisson model. That is to say, the actual variation in the claim count estimation is larger than what the Poisson GLM predicts. We'll see if the Negative Binomial model resolves the overdispersion issue.

Negative Binomial, Log-Link

As with the Poisson model, the objective for the Negative Binomial model is to

- Fit a GLM using a log-link with **Exposure** as the offset.
- Check for overdispersion at the 5% significance level.

Because Poisson and Negative Binomial GLMs are both modeling claim frequency, we can compare their relativities and AIC/BIC values side-by-side.

```
# Overdispersion test (using simulated residuals)
```

```
dispersion ratio = 0.966
p-value = < 0.001
```

Although overdispersion is not present in the Negative Binomial model, there is enough evidence to conclude, at the 5% significance level, that there is underdispersion. This time, the actual variation in the claim frequency estimation is smaller than what the Negative Binomial GLM predicts.

	PoissonRelativity	NBRelativity
(Intercept)	0.10	0.10
AreaB	1.06	1.06
AreaC	1.08	1.08
AreaD	1.17	1.17
AreaE	1.21	1.21
AreaF	1.25	1.25
VehBrandB10	1.02	1.03
VehBrandB11	1.10	1.10
VehBrandB12	1.13	1.14
VehBrandB13	1.04	1.04
VehBrandB14	0.83	0.83
VehBrandB2	0.99	0.99
VehBrandB3	1.01	1.01
VehBrandB4	0.97	0.97
VehBrandB5	1.08	1.08
VehBrandB6	0.97	0.97
VehGasRegular	1.07	1.07
DrivAgeBand26-35	0.81	0.81
DrivAgeBand36-45	1.04	1.04
DrivAgeBand46-55	1.20	1.20
DrivAgeBand56-65	1.05	1.04
DrivAgeBand66-75	1.10	1.09
DrivAgeBand75+	1.21	1.20
BonusMalusBand61-70	1.83	1.83
BonusMalusBand71-80	1.73	1.73
BonusMalusBand81-90	1.95	1.95
BonusMalusBand91-100	2.94	2.97
BonusMalusBand101-150	4.95	4.96
VehAgeBand2-4	0.59	0.58
VehAgeBand5-7	0.65	0.63
VehAgeBand8-10	0.62	0.61
VehAgeBand11-20	0.51	0.50
VehAgeBand21+	0.39	0.38
VehPowerBand5-7	1.07	1.08
VehPowerBand7-10	1.05	1.05
VehPowerBand10+	0.99	0.99
DensityBandLow-Medium	0.99	0.99
DensityBandMedium	0.91	0.91
DensityBandMedium-High	0.97	0.98
DensityBandHigh	0.96	0.97

Poisson vs. Negative Binomial Relativity Comparison

The relativities are agreed upon between both models, meaning that we can reach very similar conclusions with either one. Also, we can infer that the two models will not differ by that much when comparing metrics such as AIC/BIC.

```
Poisson AIC: 227780.6
Negative Binomial AIC: 227342.2
Better Model: Negative Binomial
```

Using AIC, the Negative Binomial is deemed as the better model, although Poisson performs basically just as well, enough for the difference to not be of significance. The Poisson model will be used in the final Frequency $\times$ Severity model.

Severity Modeling

We turn our attention to modeling the average claim amount. Unfortunately, the Inverse Gaussian model could not converge to a stable solution, leaving us with only the Gamma GLM to use for severity modeling.

Recall that the domain of the Gamma GLM is $(0, \infty)$. Therefore, we must restrict the training data to policies with a positive number of claims.

```
Average severity (train): 2183.367
Average severity (test): 1957.708
```

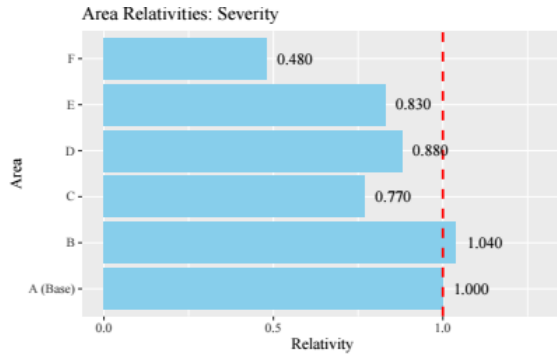
Gamma GLM, Log-Link

The primary objectives for the Gamma GLM include:

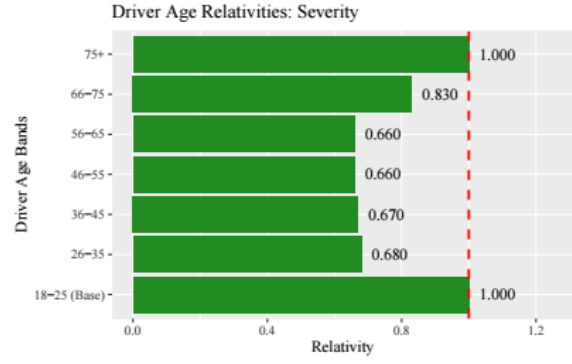
- Fit a GLM using a log-link. We do not include an offset term as we are modeling an amount, not a rate.
- Visualize severity relativities using a vertical bar graph.

Figure 10 illustrates the severity relativities fit using the Gamma model. Some interesting observations worth pointing out are:

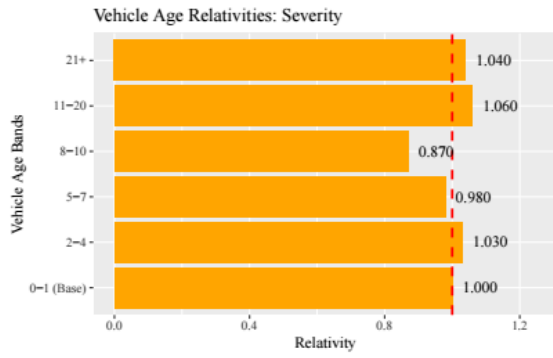
- 75+ year-old drivers have the same relativity as 18-25 year-old drivers. All other age groups have a smaller expected severity.
- Older vehicles have a slightly higher expected severity, despite having a smaller expected claim frequency.
- Policies with a **BonusMalus** of 71-100 expect significantly larger claim amounts.



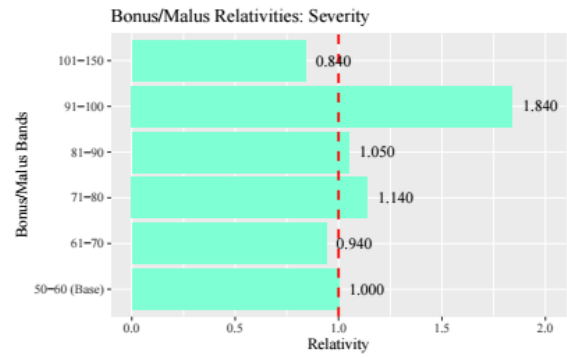
(a) Area



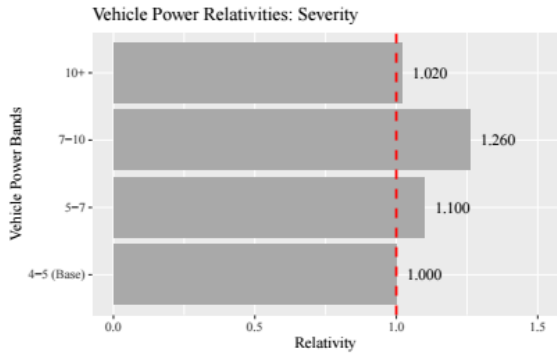
(b) DrivAgeBand



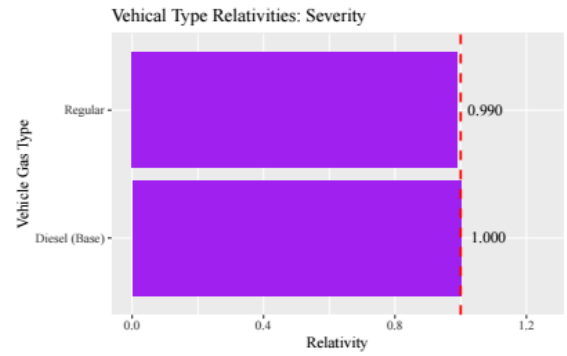
(c) VehAgeBand



(d) BonusMalusBand



(e) VehPowerBand



(f) VehGas

Figure 10: Gamma Frequency Relativities by Level

- Vehicles with a higher power have a larger expected severity.

More importantly, severity tells a completely different story than frequency. Frequency relativities do not serve as indicators for severity relativities.

Tweedie Distribution

Recall that the Tweedie Distribution estimates the pure premium directly; only one model is to be constructed as opposed to two. Also, the choice of the power in the variance relationship is not chosen arbitrarily. Typically, one fits multiple Tweedie models of powers ranging between 1 and 2, and then selecting the model with the best performance. That could be through the lowest deviance, AIC/BIC, or through other similar metrics. For the sake of this project, deviance will be used to select the best Tweedie model.

Deviance Statistic Comparisons at varying Tweedie powers

Power = 1.1: 204173257

Power = 1.3: 59575433

Power = 1.5: 21894259

Power = 1.6: 14541260

Best power: 1.6

The Tweedie models with the powers 1.7 and 1.9 were also tested, but failed to converge. Therefore, the model with power 1.6 will be used in pure premium estimations because it has the lowest deviance.

Pure Premium: Relativity and Comparison

We have built the foundation for estimating the pure premium, leaving us with a comparison between two final models:

- A Frequency $\times$ Severity model using our Poisson model for frequency and Gamma model for severity. And;
- The Tweedie model with power 1.6.

There are a few important metrics that we can now calculate with two built models:

- **Portfolio-Level Pure Premium Estimation:** Frequency $\times$ Severity vs. Tweedie. The estimation will be compared to the actual portfolio-level pure premium. This will tell us how well our models are accurately estimating the pure premium at the global level.

- **Weighted Mean Absolute Error:** Measures how far our models' predicted costs are from actual claims, assigning weights to policies' exposure lengths. This ensures that each policies with greater exposure is given more weight in the error sum and have a larger impact on the error. A lower MAE indicates more accurate predictions on individual policies on average.

Test Set - Portfolio Level Comparison:

Actual total premium amount: 11184901

Predicted FreqxSev amount: 14966252 , Ratio of actual: 1.338076

Predicted Tweedie amount: 10895852 , Ratio of actual: 0.9741573

MAE: FreqxSev: 342.1824

MAE: Tweedie: 289.1991

- The Frequency $\times$ Severity model predicted a portfolio-level pure premium amount that is 33.8% higher than the true portfolio amount, significantly overpricing the portfolio. This is possibly attributed to overestimation on either the Poisson or Gamma model as well as their multiplicative effect on the pure premium.
- The Tweedie model, on the other hand, predicted a portfolio-level pure premium amount that is 2.6% smaller than the true portfolio amount, suggesting it is well calibrated in estimating at the portfolio level.
- The Tweedie model, as seen by its lower Weighted Mean Absolute Error compared to the Frequency $\times$ Severity model, has better policy-level prediction accuracy.

One could conclude that the Tweedie distribution outperforms the Frequency $\times$ Severity model under these circumstances. However, a Frequency $\times$ Severity model remains valuable to interpreting rating factors separately. Here, the decision of using which model is reduced to a need for predictive accuracy (Tweedie) versus interpretability (Frequency $\times$ Severity).

Now, we take a look at the pure premium on individual rating factors. Figure 11 highlights how the Pure Premium compares between multiple rating factors.

- The Frequency $\times$ Severity and Tweedie estimations move in the same direction as the observed pure premiums. For example, in [DrivAge](#), the observed pure premium peaks at "18-25," declines, and maintains a relatively stable amount for the remaining levels. Both models do just that.
- Frequency $\times$ Severity is slightly closer to the observed pure premium amount than Tweedie, but basically comparable. The Tweedie model has a better grasp on individual policies as seen by its smaller weighted MAE, but Frequency $\times$ Severity has an advantage in grouped categories.
- The observed pure premium undergoes more variability and fluctuations whereas the two models provide stable estimates.

- The two models severely underestimate extreme categories. Some examples include **Area D**, **BonusMalus 81-90**, and **Region B2**.

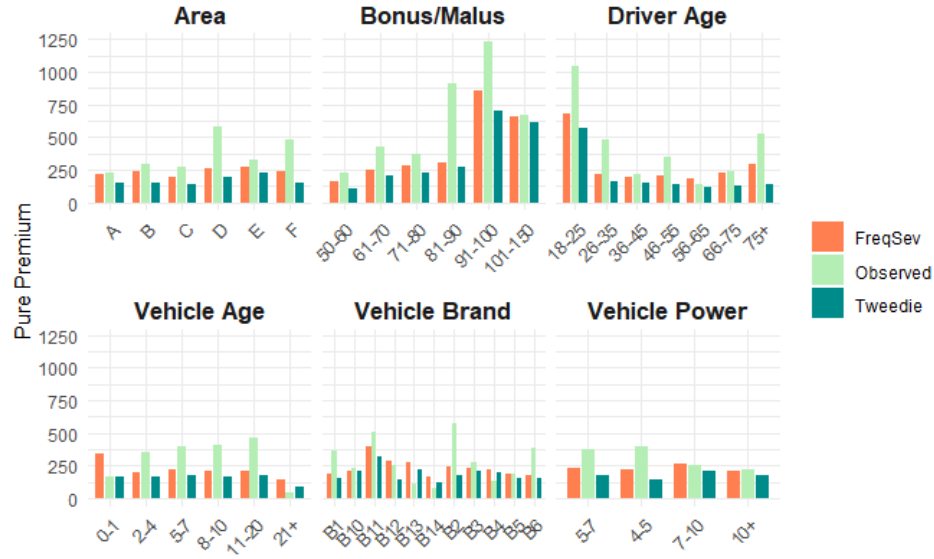


Figure 11: Categorical Rating Factors' Pure Premiums by Level

Rating Tables

Rating tables are incredibly useful to help guide pricing pure premiums. They calculate the final premium based on rating factors, thereby linking them directly to a price. Also, they ensure the insured are paying a fair amount for their coverage. For instance, we will look at the rating tables for our Poisson model.

Base Frequency: 0.09839978

Figure 12 highlights the Poisson rating table, which is to be used in conjunction with the base frequency. Note that the levels, “*, **, ***” corresponds to p -values of < 0.05 , < 0.01 , and < 0.001 , respectively. For instance, for the **DrivAgeBand** rating factor, the relativities for ages 36-45 and 56-65 are not statistically significant. However, every level in **BonusMalusBand** has a p -value less than 0.001, implying its importance in predicting claim frequency. The **Region** rating factor, on the other, hand, has fewer statistically significant levels.

As an example, a 70-year old driver with a vehicle of 18 years, BonusMalus of 50, vehicle power of 8, from area D, and their vehicle brand is B13. Lastly, the exposure period is 1 year. Then, the expected claim frequency, using the rating tables in Figure 12, is

$0.0984 \times 1.1006 \times 0.5053 \times 1.0000 \times 1.0526 \times 1.1682 \times 1.0404 \approx 0.07$ claims per year

	Relativity	Signif
(reference level)	1.0000	
26-35	0.8149	***
36-45	1.0402	
46-55	1.2015	***
56-65	1.0518	.
66-75	1.1006	**
75+	1.2148	***

(a) **DrivAgeBand**

	Relativity	Signif
(reference level)	1.0000	
2-4	0.5946	***
5-7	0.6459	***
8-10	0.6183	***
11-20	0.5053	***
21+	0.3942	***

(b) **VehAgeBand**

	Relativity	Signif
(reference level)	1.0000	
61-70	1.8324	***
71-80	1.7308	***
81-90	1.9474	***
91-100	2.9353	***
101-150	4.9506	***

(c) **BonusMalusBand**

	Relativity	Signif
(reference level)	1.0000	
5-7	1.0747	***
7-10	1.0526	**
10+	0.9922	

(d) **VehPowerBand**

	Relativity	Signif
(reference level)	1.0000	
B	1.0592	*
C	1.0825	***
D	1.1682	***
E	1.2130	***
F	1.2509	.

(e) **Area**

	Relativity	Signif
(reference level)	1.0000	
B10	1.0230	
B11	1.0985	*
B12	1.1259	***
B13	1.0404	
B14	0.8279	*
B2	0.9926	
B3	1.0089	
B4	0.9672	
B5	1.0808	**
B6	0.9696	

(f) **VehBrand**

Figure 12: Poisson Rating Tables by Categorical Rating Factor

Figure 13 offers a rating comparison between the Poisson, Gamma, and Tweedie models.

- Rating factors such as **Area**, **DrivAge**, and **VehBrand** see relativities remain clustered close to 1, and the differences between each model is small, implying that they do not contribute much to pure premium differentiation.

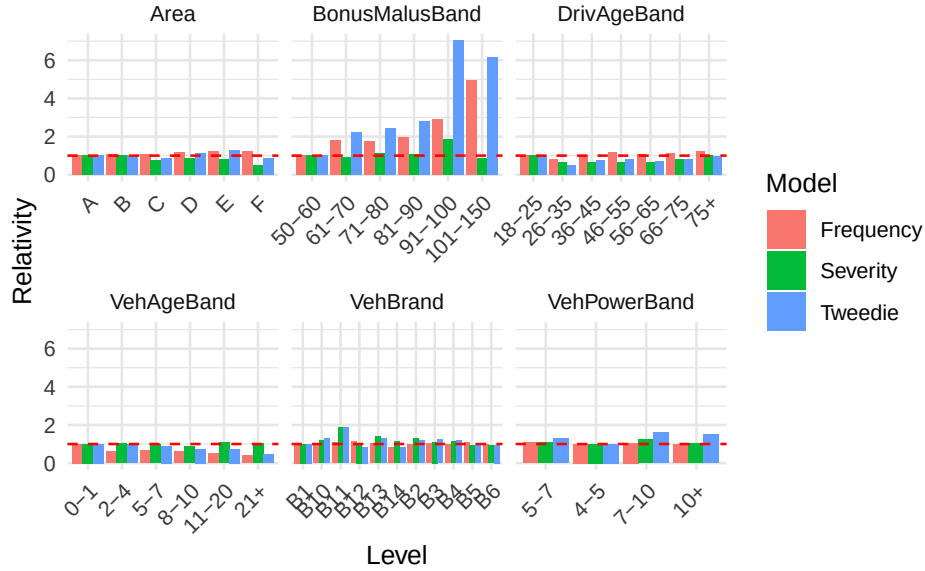


Figure 13: Rating Tables for Selected GLMs

- In **VehPower**, there is some variation in higher power categories in the Tweedie model; however, the ratings are still consistent.
- In **VehAge**, the relativities gradually decline as vehicle age increases, which is consistent among all models.
- **BonusMalus** is undoubtedly the strongest rating factor. All three models exhibit increasing relativities as the Bonus/Malus increases. There are increases in both frequency and severity (namely 91-100); nonetheless, we can confirm that the effect of **BonusMalus** on frequency and severity is substantial, as well as their combined effect on the pure premium (Tweedie).

Rating tables help accelerate the pricing structure due to its multiplicative effect; pure premiums are obtained by multiplying the corresponding GLM relativities to each rating factor. We motivate this concept with an example policy.

The example policy has the following characteristics: 45 year-old driver, BonusMalus 60, VehAge 8 years, VehPower 5, Area E, VehBrand B12, Medium Density, Diesel.

Expected Annual Frequency: 0.08437035 claims per year
Expected Average Severity: 1533.956
Expected Pure Premium: 129.4204

Model Diagnostics

A model is only as good as the assumptions imposed onto them. Here we introduce key diagnostics for GLMs that will be assessed on the Poisson frequency and Gamma severity models.

Residuals

- **Residual vs. Fitted plot:** As the diagnostic suggests, we look at a plot of the model's deviation from the observed value (y -axis) over the actual model predictions (x -axis). This gives us insight on how far observed points are from the model's predictions. A common desirable residual vs. fitted plot is a "cloud" of points that are centered around $y = 0$, implying that the model is consistently close to predicting the observed values. This is a rare feat for GLMs because of their variance structure; they are a function of the mean and non-constant, resulting in heteroscedasticity or unique residual patterns. Generally, systematic patterns in the plot is not a good sign.
- **QQ-plots:** The QQ-plot is another diagnostic that tells us if the model is adequately capturing the relationship between the rating factors and the response (claim frequency or severity). This is achieved by checking whether the model residuals (or deviances) are consistent with a "theoretical" distribution we are supposing. In a perfect case, the points on the QQ-plot follow a 45-degree line, because the sample points are matching the points on this theoretical distribution.
- **Residual Distribution:** Tells us how the prediction residuals are distributed. Ideally, the distribution is centered around zero because it tells us the model is unbiased. In the context of insurance claims, we would conclude that the model neither overestimates nor underestimates claim frequency (or severity for the Gamma model).

We employ these plots for both Poisson and Gamma GLMs. For the QQ-plot, we assume a normal distribution (mean = 0, variance = 1) as the theoretical quantiles.

Figures 14 and 15 illustrate each of the model diagnostics applied to our Poisson model.

- Figure 14 outlines the residual behavior in the Poisson model. Typically we want to see an equal distribution about $y = 0$, but the Poisson residuals look like a series of curved bands. Recall that a Poisson model predicts a discrete response variable: a count. This makes the deviance residual a deterministic function of the observed count and the fitted mean response.
- In Figure 14, the lowest band corresponds to claim-free policies, the next band corresponds to 1 claim, and so on. Most of the policies have zero claims, which is why we see many negative residuals (below dashed line). With respect to auto insurance claims, most insured drivers are claim-free during the exposure period, resulting in the low-frequency nature of auto-insurance claims.

- Positive residuals correspond to policies with claims. This means that a small number of policies produced more claims than expected.

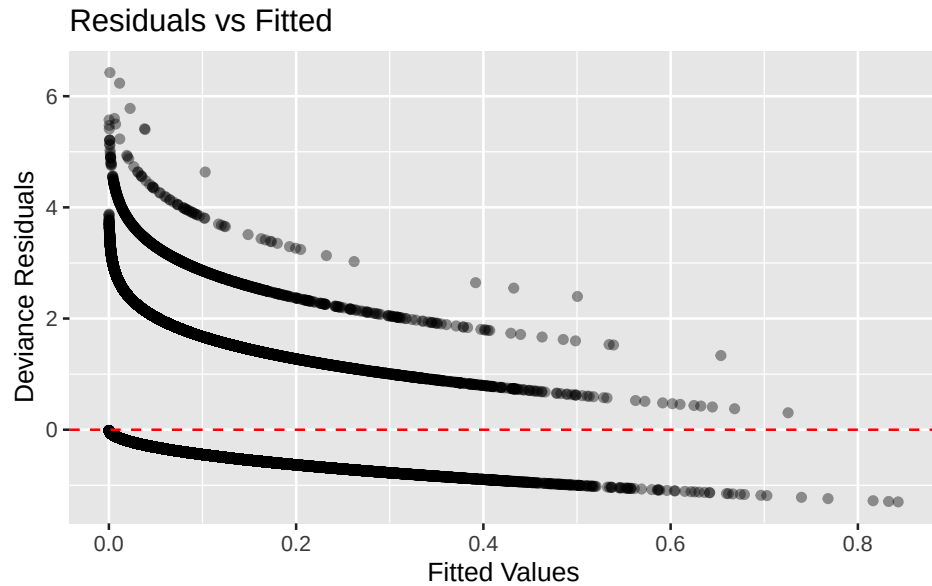


Figure 14: Poisson Residuals vs. Fitted Plot

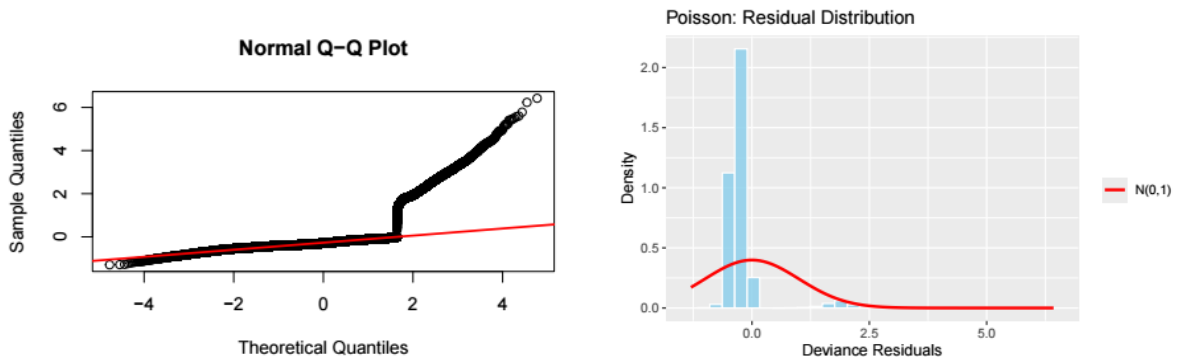


Figure 15: Poisson QQ-plot and Residual Distribution

- As the fitted value increases, the deviance residual decreases. The Poisson model assigns larger prediction errors to unexpected claims for “low-risk” policies than “high-risk” ones, highlighting noticeable risk differentiation.

- Together, the plots in Figure 15 tell us a story. From the distribution plot, we can conclude there are many small residuals and a few large positive ones. Therefore, the right-tail in the QQ-plot is a lot steeper than a normal distribution.
- In summary, our data has a large majority of claim-free policies. In our case, we have residuals clustered slightly below zero. With a smaller fraction of policies producing at least one claim, we obtain a right-skewed tail. Therefore, the distribution of the deviance residuals has a heavier right tail than a normal distribution. These extreme residuals represent policies whose claim frequency was noticeably different than the model's prediction. This is the result of unusually high-risk drivers or random claims.

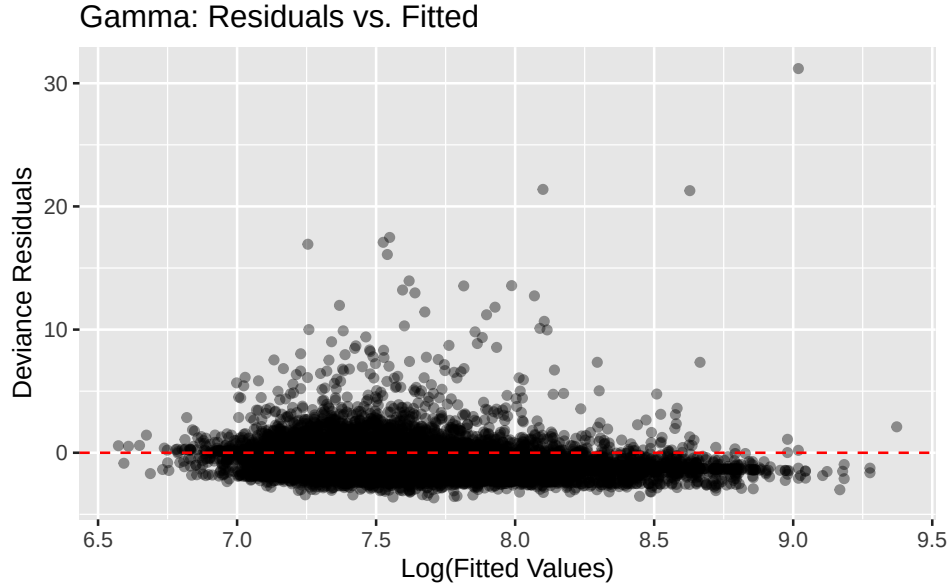


Figure 16: Gamma Residuals vs. Fitted Plot

- In Figure 16, most observations are concentrated around zero; therefore, the Gamma GLM fits the majority of policies well. There is also no strong curvature or systematic patterns in the residuals, suggesting that the log-link specification is appropriate.
- There are a handful of large positive residuals, which can be attributed to the Gamma GLM underestimating some high-severity claims.
- The residual spread does narrow slightly as the fitted values increase, indicating a mild form of heteroscedasticity. As discussed in the Methodology section, this behavior is consistent with the variance structure of the Gamma distribution.

Figure 17 displays the QQ-plot and residual distribution for the Gamma GLM.

- The QQ-plot on the left-hand panel illustrates a noticeable deviation from normality in the upper tail, indicating the existence of extreme claim amounts. We were able to catch a glimpse of this when looking at the statistical summary for the `Claim.Amount` variable. This does not indicate poor performance of the GLM, however, especially if there are a few claims that overwhelmingly dominate the tail.

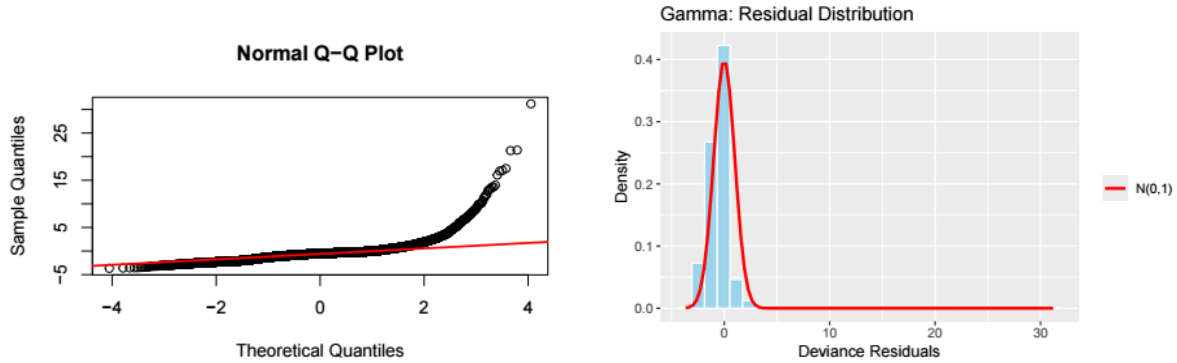


Figure 17: Gamma QQ-plot and Residual Distribution

- In the right-hand panel, we notice that the residual distribution is mostly centered about zero, and that there is a right-tail extending beyond 30. Therefore, the normal distribution does not match the observed distribution.

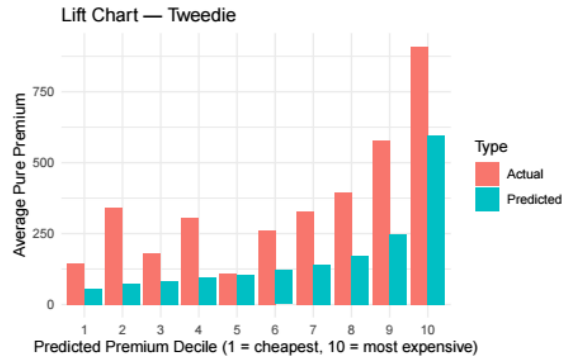
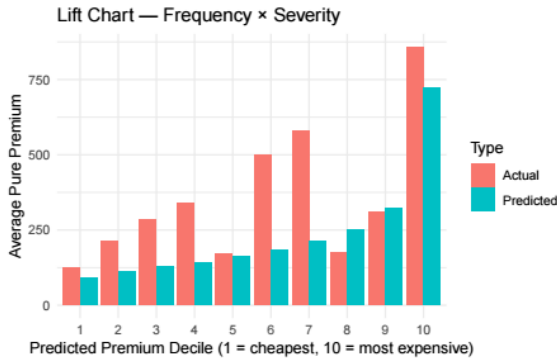
Lift Charts

Next, we look at a lift chart of the Frequency $\times$ Severity and Tweedie models. A lift chart, in our case, will partition policies into deciles by the predicting premium. From there, we want to see if the actual average severity increases from the cheapest to the most costly decile. Another key indicator of performance is measuring the absolute difference between the predicted and actual premium; the smaller the better.

Lift charts are a powerful tool to determine whether low-risk policies are high-risk are correctly identified and isolated.

Figure 18: Frequency $\times$ Severity and Tweedie Lift Charts

- There is a general increase in actual premium for both models.
- In both models, the predicted premiums monotonically increase over each decile, which is a good indication of ranking policy risk.



- The Frequency $\times$ Severity model suffers from a large deviation in the 6th and 7th deciles, underpredicting the observed average premium. Otherwise, there is a closer representation for the other deciles.
- The Tweedie model almost always underpredicts the observed premium, even more so in higher-risk policies.
- Some of these jumps can be attributed to the extreme claim amounts we have seen; they can raise the average claim amount significantly and produce these gaps in actual vs. predicted premiums for each decile. Therefore, we cannot necessarily imply that either model is expected to perform worse.

Lorenz Curves and Gini Coefficients

Much like lift charts, Lorenz curves and Gini coefficients, when combined, are an extremely effective standard for evaluating pricing models. They test how well the model differentiates low- vs. high-risk policies. Below is a summary of how Lorenz curves work:

- The x -axis represents the cumulative percentage of exposure and the y -axis represents the cumulative percentage of claim severity, both on the test data. Because we are plotting an ordered Lorenz curve, the policies are ordered according to their predicted pure premiums.
- A 45-degree line implies that the model is random: each policy has the same risk. This is the worst-case scenario, because this limits our ability to price according to specific rating factors.
- In an ideal pricing model, we want a steep curve, or maximum separation from the 45-degree line. This is because the model would be able to correctly identify all of the low-risk policies first. For example, in a L-shaped Lorenz curve, all claims are concentrated in the highest-risk policies. This corresponds to perfect separation.

The Gini coefficient is equal to twice the area between the Lorenz curve and the 45-degree line. The closer this quantity is to 1, the better.

Figure 19 illustrates the Lorenz Curves for both models.

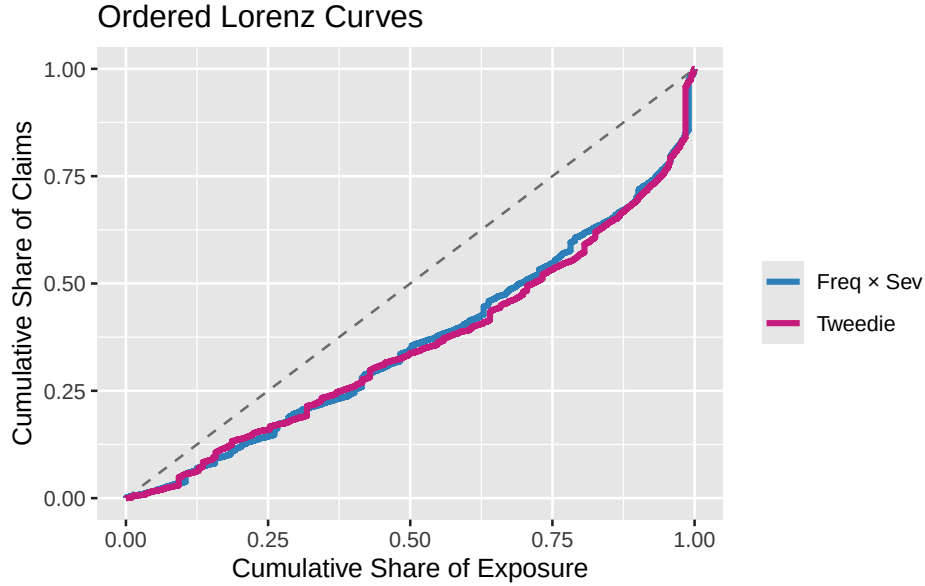


Figure 19: Lorenz Curve for Frequency $\times$ Severity and Tweedie Models

FreqxSev Gini: 0.2773341

Tweedie (p = 1.6) Gini: 0.2872465

- The two curves practically overlap, implying that the risk separation is very similar between the Frequency $\times$ Severity and Tweedie models.
- The Tweedie curve's deviation is slightly farther than Frequency $\times$ Severity, suggesting slightly better risk inequality. This is reinforced when comparing the two Gini coefficients: the Tweedie model's Gini coefficient is roughly 0.01 larger than the Frequency $\times$ Severity model (0.2872 vs. 0.2773). This difference is quite small; the two models perform similarly in ranking policies by risk.
- Note that a Gini coefficient of 0.2872, while far from 1, does not imply poor separation. Both models have a moderate ability to separate low- and high-risks correctly, although neither achieve perfect separation.

Model Comparison and Summary

In this section, we will:

- Provide GLM properties for the following models: Poisson, Negative Binomial, Gamma, Tweedie (for power of 1.6). Such properties include the mean response variable, family, link function, and number of parameters.
- For each fitted GLM, compare Log-likelihood, deviance, Pearson chi-squared, AIC, and BIC under their corresponding degrees of freedom. Models that could not retrieve the following values will have a "-."
- For each fitted GLM, compare dispersion parameters and scale.
- For each model, compare deviance, MAE, and RMSE on the test set.
- For each model, compare portfolio rate and weighted MAE on the test set.
- Conclude the report with some closing remarks.

GOODNESS-OF-FIT METRICS - MODEL COMPARISON

	Model	Target	Family	Link	NumParams
1	Poisson	Frequency	poisson	log	40
2	Gamma	Severity	Gamma	log	40
3	Tweedie (p=1.6)	Pure Premium	Tweedie	mu ⁰	40

	Model	LogLikelihood	Deviance	PearsonChi2
1	Poisson	-113850.3	172310.2	1317020
2	Gamma	-170514.6	31436.9	614018
3	Tweedie (p=1.6)	-	14541260.3	3411435490

	Model	AIC	BIC	DfResidual	DfModel
1	Poisson	227780.6	228228.7	540984	39
2	Gamma	341111.2	341435	19849	39
3	Tweedie (p=1.6)	-	-	540984	39

	Model	Dispersion_Pearson	Dispersion_Deviance	Scale
1	Poisson	2.434490	0.3185126	1.00000
2	Negative Binomial	2.358114	0.2839052	1.00000
3	Gamma	30.934456	1.5838005	30.93446
4	Tweedie (p=1.6)	6305.982229	26.8792798	6307.56626

```

=====
TEST SET PERFORMANCE
=====
--- Frequency Model (Test Set) ---
          Deviance      MAE      RMSE
Poisson    172310.2    0.0985    0.2367
--- Severity Model (Test Set, claims > 0) ---
          Deviance      MAE      RMSE
Gamma      31436.9    2,027    4,426
--- Pure Premium Model (Test Set) ---
          Portfolio Ratio    Weighted MAE
Tweedie          0.9742          289

          Model      Target      Distribution    AIC_Train    BIC_Train
          Poisson    Frequency    Poisson    227780.6    228228.7
Negative Binomial    Frequency    Negative Binomial    227342.2    227801.4
          Gamma      Severity      Gamma    341111.2    341435
          Tweedie (p=1.6) Pure Premium    Tweedie (p=1.6)    -    -
          FreqxSev Pure Premium    Poisson x Gamma    -    -
LogLik_Train Pearson_Dispersion
-113850.3          2.4
-113630.1          2.4
-170514.6        30.9
-          6306
-          -

```

- Each model offered a reasonable fit to the policy data; residual diagnostics showed that the selected distributions and log-links sufficiently captured the overall claim behavior, despite the presence of extreme outliers and therefore a heavy right-skewed tail.
- The candidate models for measuring average pure premiums, Frequency $\times$ Severity and Tweedie, displayed comparable performance on the test data, with slight differences in the performance metrics.
- Residual diagnostics, lift charts, and ordered Lorenz curves consistently proved similar the ability to measure risk inequality between policies, even though the Tweedie model had a slightly higher Gini coefficient, thereby suggesting a modest increase in risk differentiation.
- Each of the two candidate models has an advantage in pricing. The Frequency $\times$ Severity model allows us to gather more information about the impact of rating factors on claim frequency and severity, whereas the Tweedie distribution only requires the construction of a single model for measuring the average pure premium.

- Pricing models are the result of the strength between rating factors and the mean response. Exploratory data analysis showed that **BonusMalus** is a very significant rating factor in identifying patterns in claim frequency and severity.

In summary, both models performed very similarly on all fronts—predicting the average pure premium as well as risk segmentation—with the Tweedie model having a slight edge over Frequency $\times$ Severity. Therefore, the model to be chosen for future pricing depends on the application. Using a Frequency $\times$ Severity model offers a more interactive understanding of the policy data; we can obtain cross-sectional analysis on rating factor relativities for both claim frequency and severity, allowing us to identify potential patterns. However, if predictability and simplicity is the objective, then the Tweedie model is the more suitable choice, as we can obtain estimates on the average pure premium directly.

We briefly provide examples of when these advantages can be understood from a business perspective:

- Underwriters may need to justify rate changes, which can be due to poor driving habits (frequency) or medical and cost inflation (severity). The Frequency $\times$ Severity model provides this cross-sectional analysis.
- On the other hand, the Tweedie model makes the rate-changing process easier by only needing to defend a single model instead of two. The model itself is elegantly handled for autocalibration, forcing total predicted premiums to match total actual premiums at the portfolio level. A Frequency $\times$ Severity would have twice the statistical assumptions and model multiplication is a harder step to justify, particularly because autocalibration is nearly impossible to achieve this way.

Please note that this report is not representative of a complete analysis on pricing for this dataset. There are some additions and changes that can be done to further polish this report:

- Use resampling techniques such as k -fold cross-validation on predictive metrics (weighted MAE, RMSE, and AIC/BIC)
- Plot the deviance over Tweedie powers and find the minimum.
- Model large claims separately (ex. any policies with **Claim.Amount** above 25000)
- Run ANOVA tests to identify if only a particular subset of rating factors contributes to lower AIC or deviance.

Appendix

Frequency Modeling

Poisson GLM Summary

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-2.318717	0.036124	-64.188	< 2e-16	***
AreaB	0.057516	0.023999	2.397	0.01655	*
AreaC	0.079275	0.019425	4.081	4.48e-05	***
AreaD	0.155443	0.020145	7.716	1.20e-14	***
AreaE	0.193081	0.021726	8.887	< 2e-16	***
AreaF	0.223889	0.124547	1.798	0.07224	.
VehBrandB10	0.022726	0.041415	0.549	0.58318	
VehBrandB11	0.093985	0.044371	2.118	0.03416	*
VehBrandB12	0.118626	0.019462	6.095	1.09e-09	***
VehBrandB13	0.039630	0.046029	0.861	0.38925	
VehBrandB14	-0.188810	0.089409	-2.112	0.03471	*
VehBrandB2	-0.007389	0.017169	-0.430	0.66693	
VehBrandB3	0.008865	0.024476	0.362	0.71720	
VehBrandB4	-0.033310	0.033541	-0.993	0.32066	
VehBrandB5	0.077670	0.027709	2.803	0.00506	**
VehBrandB6	-0.030860	0.031837	-0.969	0.33239	
VehGasRegular	0.063444	0.012454	5.094	3.50e-07	***
DrivAgeBand26-35	-0.204688	0.026569	-7.704	1.32e-14	***
DrivAgeBand36-45	0.039373	0.027503	1.432	0.15226	
DrivAgeBand46-55	0.183567	0.027893	6.581	4.67e-11	***
DrivAgeBand56-65	0.050549	0.030338	1.666	0.09567	.
DrivAgeBand66-75	0.095863	0.033383	2.872	0.00408	**
DrivAgeBand75+	0.194591	0.040457	4.810	1.51e-06	***
BonusMalusBand61-70	0.605619	0.020641	29.341	< 2e-16	***
BonusMalusBand71-80	0.548569	0.022108	24.813	< 2e-16	***
BonusMalusBand81-90	0.666500	0.027077	24.615	< 2e-16	***
BonusMalusBand91-100	1.076819	0.024786	43.445	< 2e-16	***
BonusMalusBand101-150	1.599506	0.034205	46.762	< 2e-16	***
VehAgeBand2-4	-0.519847	0.018370	-28.298	< 2e-16	***
VehAgeBand5-7	-0.437177	0.020385	-21.446	< 2e-16	***
VehAgeBand8-10	-0.480772	0.021160	-22.721	< 2e-16	***
VehAgeBand11-20	-0.682557	0.020017	-34.100	< 2e-16	***
VehAgeBand21+	-0.930826	0.071866	-12.952	< 2e-16	***
VehPowerBand5-7	0.071999	0.013683	5.262	1.43e-07	***

VehPowerBand7-10	0.051284	0.018808	2.727	0.00640	**
VehPowerBand10+	-0.007856	0.030969	-0.254	0.79976	
DensityBandLow-Medium	-0.010254	0.030829	-0.333	0.73943	
DensityBandMedium	-0.096130	0.151058	-0.636	0.52453	
DensityBandMedium-High	-0.027457	0.154979	-0.177	0.85938	
DensityBandHigh	-0.036533	0.130285	-0.280	0.77917	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 178636 on 541023 degrees of freedom
 Residual deviance: 172310 on 540984 degrees of freedom
 AIC: 227781

Number of Fisher Scoring iterations: 6

Poisson GLM Relativity

	Coefficient	Relativity	Std..Error	z.value	Pr...z..
(Intercept)	-2.3187	0.10	0.04	-64.19	0.00
AreaB	0.0575	1.06	0.02	2.40	0.02
AreaC	0.0793	1.08	0.02	4.08	0.00
AreaD	0.1554	1.17	0.02	7.72	0.00
AreaE	0.1931	1.21	0.02	8.89	0.00
AreaF	0.2239	1.25	0.12	1.80	0.07
VehBrandB10	0.0227	1.02	0.04	0.55	0.58
VehBrandB11	0.0940	1.10	0.04	2.12	0.03
VehBrandB12	0.1186	1.13	0.02	6.10	0.00
VehBrandB13	0.0396	1.04	0.05	0.86	0.39
VehBrandB14	-0.1888	0.83	0.09	-2.11	0.03
VehBrandB2	-0.0074	0.99	0.02	-0.43	0.67
VehBrandB3	0.0089	1.01	0.02	0.36	0.72
VehBrandB4	-0.0333	0.97	0.03	-0.99	0.32
VehBrandB5	0.0777	1.08	0.03	2.80	0.01
VehBrandB6	-0.0309	0.97	0.03	-0.97	0.33
VehGasRegular	0.0634	1.07	0.01	5.09	0.00
DrivAgeBand26-35	-0.2047	0.81	0.03	-7.70	0.00
DrivAgeBand36-45	0.0394	1.04	0.03	1.43	0.15
DrivAgeBand46-55	0.1836	1.20	0.03	6.58	0.00
DrivAgeBand56-65	0.0505	1.05	0.03	1.67	0.10
DrivAgeBand66-75	0.0959	1.10	0.03	2.87	0.00

DrivAgeBand75+	0.1946	1.21	0.04	4.81	0.00
BonusMalusBand61-70	0.6056	1.83	0.02	29.34	0.00
BonusMalusBand71-80	0.5486	1.73	0.02	24.81	0.00
BonusMalusBand81-90	0.6665	1.95	0.03	24.61	0.00
BonusMalusBand91-100	1.0768	2.94	0.02	43.44	0.00
BonusMalusBand101-150	1.5995	4.95	0.03	46.76	0.00
VehAgeBand2-4	-0.5198	0.59	0.02	-28.30	0.00
VehAgeBand5-7	-0.4372	0.65	0.02	-21.45	0.00
VehAgeBand8-10	-0.4808	0.62	0.02	-22.72	0.00
VehAgeBand11-20	-0.6826	0.51	0.02	-34.10	0.00
VehAgeBand21+	-0.9308	0.39	0.07	-12.95	0.00
VehPowerBand5-7	0.0720	1.07	0.01	5.26	0.00
VehPowerBand7-10	0.0513	1.05	0.02	2.73	0.01
VehPowerBand10+	-0.0079	0.99	0.03	-0.25	0.80
DensityBandLow-Medium	-0.0103	0.99	0.03	-0.33	0.74
DensityBandMedium	-0.0961	0.91	0.15	-0.64	0.52
DensityBandMedium-High	-0.0275	0.97	0.15	-0.18	0.86
DensityBandHigh	-0.0365	0.96	0.13	-0.28	0.78

Negative Binomial GLM Summary

	Coefficient	Relativity	Std..Error	z.value	Pr...z..
(Intercept)	-2.2956	0.10	0.04	-61.36	0.00
AreaB	0.0576	1.06	0.02	2.34	0.02
AreaC	0.0793	1.08	0.02	3.98	0.00
AreaD	0.1561	1.17	0.02	7.54	0.00
AreaE	0.1938	1.21	0.02	8.67	0.00
AreaF	0.2267	1.25	0.13	1.76	0.08
VehBrandB10	0.0260	1.03	0.04	0.61	0.54
VehBrandB11	0.0949	1.10	0.05	2.07	0.04
VehBrandB12	0.1321	1.14	0.02	6.60	0.00
VehBrandB13	0.0393	1.04	0.05	0.83	0.41
VehBrandB14	-0.1874	0.83	0.09	-2.05	0.04
VehBrandB2	-0.0077	0.99	0.02	-0.44	0.66
VehBrandB3	0.0090	1.01	0.03	0.36	0.72
VehBrandB4	-0.0324	0.97	0.03	-0.94	0.35
VehBrandB5	0.0800	1.08	0.03	2.80	0.01
VehBrandB6	-0.0298	0.97	0.03	-0.91	0.36
VehGasRegular	0.0676	1.07	0.01	5.27	0.00
DrivAgeBand26-35	-0.2093	0.81	0.03	-7.58	0.00
DrivAgeBand36-45	0.0366	1.04	0.03	1.28	0.20

DrivAgeBand46-55	0.1814	1.20	0.03	6.24	0.00
DrivAgeBand56-65	0.0432	1.04	0.03	1.37	0.17
DrivAgeBand66-75	0.0891	1.09	0.03	2.57	0.01
DrivAgeBand75+	0.1857	1.20	0.04	4.42	0.00
BonusMalusBand61-70	0.6025	1.83	0.02	28.15	0.00
BonusMalusBand71-80	0.5488	1.73	0.02	24.06	0.00
BonusMalusBand81-90	0.6689	1.95	0.03	23.89	0.00
BonusMalusBand91-100	1.0882	2.97	0.03	42.13	0.00
BonusMalusBand101-150	1.6009	4.96	0.04	43.11	0.00
VehAgeBand2-4	-0.5421	0.58	0.02	-28.66	0.00
VehAgeBand5-7	-0.4565	0.63	0.02	-21.70	0.00
VehAgeBand8-10	-0.4980	0.61	0.02	-22.82	0.00
VehAgeBand11-20	-0.7020	0.50	0.02	-34.04	0.00
VehAgeBand21+	-0.9547	0.38	0.07	-12.99	0.00
VehPowerBand5-7	0.0731	1.08	0.01	5.18	0.00
VehPowerBand7-10	0.0474	1.05	0.02	2.45	0.01
VehPowerBand10+	-0.0137	0.99	0.03	-0.43	0.67
DensityBandLow-Medium	-0.0083	0.99	0.03	-0.26	0.79
DensityBandMedium	-0.0963	0.91	0.16	-0.62	0.54
DensityBandMedium-High	-0.0197	0.98	0.16	-0.12	0.90
DensityBandHigh	-0.0342	0.97	0.13	-0.25	0.80

Severity Modeling

Gamma GLM Summary

Call:

```
glm(formula = AvgClaimAmount ~ . - PurePremium - Exposure - ClaimNb -
     Claim.Amount, family = Gamma(link = "log"), data = train_sev)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.8492525	0.2487289	31.557	< 2e-16 ***
AreaB	0.0383183	0.1650948	0.232	0.816464
AreaC	-0.2662644	0.1335661	-1.994	0.046220 *
AreaD	-0.1240577	0.1367065	-0.907	0.364167
AreaE	-0.1888827	0.1461855	-1.292	0.196346
AreaF	-0.7409337	0.7988624	-0.927	0.353686
VehBrandB10	0.1814188	0.2546230	0.712	0.476164
VehBrandB11	0.6335815	0.2635361	2.404	0.016219 *
VehBrandB12	0.0520025	0.1409419	0.369	0.712159

VehBrandB13	0.3429070	0.2890334	1.186	0.235482
VehBrandB14	0.1133716	0.5836259	0.194	0.845979
VehBrandB2	0.2415953	0.1100794	2.195	0.028194 *
VehBrandB3	0.1021364	0.1531492	0.667	0.504839
VehBrandB4	0.1284711	0.2078225	0.618	0.536465
VehBrandB5	-0.0674213	0.1761885	-0.383	0.701972
VehBrandB6	-0.0838601	0.1972442	-0.425	0.670725
VehGasRegular	-0.0080815	0.0826673	-0.098	0.922124
DrivAgeBand26-35	-0.3890951	0.1739459	-2.237	0.025306 *
DrivAgeBand36-45	-0.3996840	0.1806607	-2.212	0.026954 *
DrivAgeBand46-55	-0.4202945	0.1839031	-2.285	0.022299 *
DrivAgeBand56-65	-0.4215529	0.2012065	-2.095	0.036172 *
DrivAgeBand66-75	-0.1843856	0.2301977	-0.801	0.423148
DrivAgeBand75+	0.0002779	0.2826587	0.001	0.999215
BonusMalusBand61-70	-0.0631650	0.1280766	-0.493	0.621890
BonusMalusBand71-80	0.1277282	0.1384137	0.923	0.356123
BonusMalusBand81-90	0.0530111	0.1721837	0.308	0.758181
BonusMalusBand91-100	0.6095655	0.1687022	3.613	0.000303 ***
BonusMalusBand101-150	-0.1721001	0.2109918	-0.816	0.414698
VehAgeBand2-4	0.0262919	0.1392969	0.189	0.850293
VehAgeBand5-7	-0.0184524	0.1489004	-0.124	0.901377
VehAgeBand8-10	-0.1440146	0.1520609	-0.947	0.343607
VehAgeBand11-20	0.0599830	0.1446402	0.415	0.678362
VehAgeBand21+	0.0422741	0.4833475	0.087	0.930306
VehPowerBand5-7	0.0934751	0.0918973	1.017	0.309085
VehPowerBand7-10	0.2279499	0.1259673	1.810	0.070374 .
VehPowerBand10+	0.0234632	0.1962816	0.120	0.904850
DensityBandLow-Medium	0.1939412	0.2024998	0.958	0.338208
DensityBandMedium	0.0734565	1.0088848	0.073	0.941958
DensityBandMedium-High	0.3506686	1.0370339	0.338	0.735257
DensityBandHigh	0.2983359	0.8422602	0.354	0.723186

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Gamma family taken to be 30.93446)

Null deviance: 35300 on 19888 degrees of freedom
Residual deviance: 31437 on 19849 degrees of freedom
AIC: 341111

Number of Fisher Scoring iterations: 25

Gamma GLM Relativity Summary

	Coefficient	Relativity	Std..Error	t.value	Pr...t..
(Intercept)	7.8493	2563.82	0.25	31.56	0.000
AreaB	0.0383	1.04	0.17	0.23	0.816
AreaC	-0.2663	0.77	0.13	-1.99	0.046
AreaD	-0.1241	0.88	0.14	-0.91	0.364
AreaE	-0.1889	0.83	0.15	-1.29	0.196
AreaF	-0.7409	0.48	0.80	-0.93	0.354
VehBrandB10	0.1814	1.20	0.25	0.71	0.476
VehBrandB11	0.6336	1.88	0.26	2.40	0.016
VehBrandB12	0.0520	1.05	0.14	0.37	0.712
VehBrandB13	0.3429	1.41	0.29	1.19	0.235
VehBrandB14	0.1134	1.12	0.58	0.19	0.846
VehBrandB2	0.2416	1.27	0.11	2.19	0.028
VehBrandB3	0.1021	1.11	0.15	0.67	0.505
VehBrandB4	0.1285	1.14	0.21	0.62	0.536
VehBrandB5	-0.0674	0.93	0.18	-0.38	0.702
VehBrandB6	-0.0839	0.92	0.20	-0.43	0.671
VehGasRegular	-0.0081	0.99	0.08	-0.10	0.922
DrivAgeBand26-35	-0.3891	0.68	0.17	-2.24	0.025
DrivAgeBand36-45	-0.3997	0.67	0.18	-2.21	0.027
DrivAgeBand46-55	-0.4203	0.66	0.18	-2.29	0.022
DrivAgeBand56-65	-0.4216	0.66	0.20	-2.10	0.036
DrivAgeBand66-75	-0.1844	0.83	0.23	-0.80	0.423
DrivAgeBand75+	0.0003	1.00	0.28	0.00	0.999
BonusMalusBand61-70	-0.0632	0.94	0.13	-0.49	0.622
BonusMalusBand71-80	0.1277	1.14	0.14	0.92	0.356
BonusMalusBand81-90	0.0530	1.05	0.17	0.31	0.758
BonusMalusBand91-100	0.6096	1.84	0.17	3.61	0.000
BonusMalusBand101-150	-0.1721	0.84	0.21	-0.82	0.415
VehAgeBand2-4	0.0263	1.03	0.14	0.19	0.850
VehAgeBand5-7	-0.0185	0.98	0.15	-0.12	0.901
VehAgeBand8-10	-0.1440	0.87	0.15	-0.95	0.344
VehAgeBand11-20	0.0600	1.06	0.14	0.41	0.678
VehAgeBand21+	0.0423	1.04	0.48	0.09	0.930
VehPowerBand5-7	0.0935	1.10	0.09	1.02	0.309
VehPowerBand7-10	0.2279	1.26	0.13	1.81	0.070
VehPowerBand10+	0.0235	1.02	0.20	0.12	0.905
DensityBandLow-Medium	0.1939	1.21	0.20	0.96	0.338
DensityBandMedium	0.0735	1.08	1.01	0.07	0.942
DensityBandMedium-High	0.3507	1.42	1.04	0.34	0.735

DensityBandHigh	0.2983	1.35	0.84	0.35	0.723
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Pure Premium: Relativity and Comparison

Relativity Comparison Table

	PoissonRelativity	GammaRelativity	FreqxSev
(Intercept)	0.10	2563.82	252.28
AreaB	1.06	1.04	1.10
AreaC	1.08	0.77	0.83
AreaD	1.17	0.88	1.03
AreaE	1.21	0.83	1.00
AreaF	1.25	0.48	0.60
VehBrandB10	1.02	1.20	1.23
VehBrandB11	1.10	1.88	2.07
VehBrandB12	1.13	1.05	1.19
VehBrandB13	1.04	1.41	1.47
VehBrandB14	0.83	1.12	0.93
VehBrandB2	0.99	1.27	1.26
VehBrandB3	1.01	1.11	1.12
VehBrandB4	0.97	1.14	1.10
VehBrandB5	1.08	0.93	1.01
VehBrandB6	0.97	0.92	0.89
VehGasRegular	1.07	0.99	1.06
DrivAgeBand26-35	0.81	0.68	0.55
DrivAgeBand36-45	1.04	0.67	0.70
DrivAgeBand46-55	1.20	0.66	0.79
DrivAgeBand56-65	1.05	0.66	0.69
DrivAgeBand66-75	1.10	0.83	0.92
DrivAgeBand75+	1.21	1.00	1.22
BonusMalusBand61-70	1.83	0.94	1.72
BonusMalusBand71-80	1.73	1.14	1.97
BonusMalusBand81-90	1.95	1.05	2.05
BonusMalusBand91-100	2.94	1.84	5.40
BonusMalusBand101-150	4.95	0.84	4.17
VehAgeBand2-4	0.59	1.03	0.61
VehAgeBand5-7	0.65	0.98	0.63
VehAgeBand8-10	0.62	0.87	0.54
VehAgeBand11-20	0.51	1.06	0.54
VehAgeBand21+	0.39	1.04	0.41
VehPowerBand5-7	1.07	1.10	1.18
VehPowerBand7-10	1.05	1.26	1.32

VehPowerBand10+	0.99	1.02	1.02
DensityBandLow-Medium	0.99	1.21	1.02
DensityBandMedium	0.91	1.08	0.98
DensityBandMedium-High	0.97	1.42	1.38
DensityBandHigh	0.96	1.35	1.30
TweedieRelativity			
(Intercept)	124.70		
AreaB	1.00		
AreaC	0.87		
AreaD	1.11		
AreaE	1.27		
AreaF	0.88		
VehBrandB10	1.27		
VehBrandB11	1.87		
VehBrandB12	0.84		
VehBrandB13	1.30		
VehBrandB14	0.83		
VehBrandB2	1.21		
VehBrandB3	1.22		
VehBrandB4	1.19		
VehBrandB5	1.04		
VehBrandB6	0.95		
VehGasRegular	0.86		
DrivAgeBand26-35	0.51		
DrivAgeBand36-45	0.74		
DrivAgeBand46-55	0.81		
DrivAgeBand56-65	0.70		
DrivAgeBand66-75	0.80		
DrivAgeBand75+	0.95		
BonusMalusBand61-70	2.21		
BonusMalusBand71-80	2.46		
BonusMalusBand81-90	2.80		
BonusMalusBand91-100	7.06		
BonusMalusBand101-150	6.19		
VehAgeBand2-4	0.97		
VehAgeBand5-7	0.90		
VehAgeBand8-10	0.74		
VehAgeBand11-20	0.74		
VehAgeBand21+	0.45		
VehPowerBand5-7	1.28		
VehPowerBand7-10	1.62		
VehPowerBand10+	1.51		
DensityBandLow-Medium	1.10		

DensityBandMedium	1.16
DensityBandMedium-High	1.02
DensityBandHigh	0.79